CODESIGN

LAB 1 : OPENCL Programming

Rq: *The **report** should be concise and rely primarily on **figures**, and **visual explanations**. Keep the written text to a minimum. Avoid using ChatGPT !!*

1. Give the characteristics of the opencl compatible devices installed on your PC: device reference, global, local, cache Memories sizes, Number of compute units (Streaming MultiProcessors), max number of work items, work groups, etc….

A- Matrix Multiplication Kernel Optimization :

1- Starting from the performances, obtained by the C***oalsced kernel'***, as a reference value:

- Propose improvements that may increase the matrix multiplication performance (minimum 2 improvements).
- For **each** of the proposed improvements:

  Explain the used technique, how and why it improves the performances. Implement the corresponding kernel in OpenCL and give the performance gain obtained

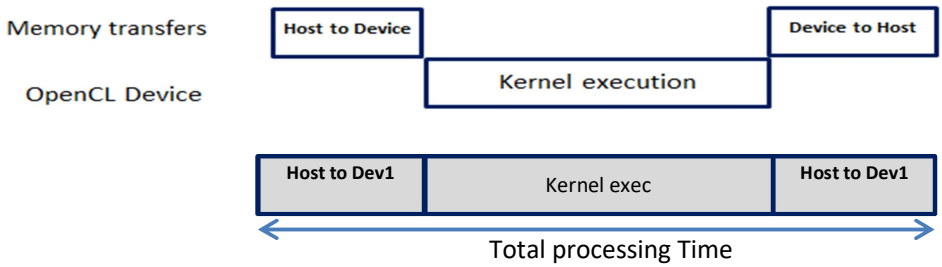
  (= Proposed technique performance / 'COALSCED' performance)

Recommended **reading:** **https://cnugteren.github.io/tutorial/pages/page1.html**

" Tutorial: OpenCL SGEMM tuning "

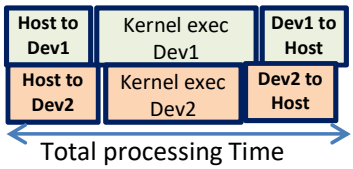
B- Running the kernel on multiple OpenCL devices

When running a kernel on a OpenCL device, the total processing time is the durations sum of data transfers and kernel execution:



To improve the performance on a PC equipped with many **OpenCL** devices (3 generally: the CPU, the integrated GPU and the dedicated GPU), it is possible to split the processing (matrix multiplication in this case) on the 3 devices and execute the kernel instances in parallel (see fig below).

In what follows, we will only use 2 opencl devices: the 2 GPUs (The integrated one and the NVIDIA dedicated GPU)



Our goal is to speed up the processing of the **UNCOALSCED** implementation on the dedicated GPU (NVIDIA) for **N=8192** and a **work-group size = 16*16**:

- The NVIDIA will process a part of the matrix C (using uncoalsced Access)
- The integrated GPU will process the remaining part of matrix C (using the method with the best performance found in "**A**").

The speedup will be evaluated:

Speedup = Performance in GFLOPS/sec (2 OpenCL devices) / Performance (NVIDIA)

1- Give the performances of the 2 GPUs for the whole Matrix multiplication (8192).

GPU	Method	Performance (GFLOPS/sec)
NVIDIA	UNCOALSCED	……
Integrated	Technique with Best performance (from part B)	…….

2- From the results found **1**) :

Explain how to split the Matrix Multiplication on the 2 devices: justify the choice of ***the sub-matrices sizes*** processed by each opencl device.

3- Give the openCL Host code and the obtained results (speedup value).